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A New Data Glove Approach for Malaysian Sign Language Detection

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Problem Addressed

Communication between deaf or speech-impaired individuals and normal people is difficult because most people do not understand sign language. Therefore, a system is needed to translate sign language gestures into readable text.

Research Gap

Many existing sign language recognition systems rely on **vision-based methods**, which are affected by lighting conditions, background noise, and camera position. These limitations reduce recognition accuracy and reliability.

Key Contribution

The paper proposes a **data glove-based sign language recognition system** that uses **tilt sensors and an accelerometer** to capture finger bending and hand motion. The system translates Malaysian Sign Language gestures into text displayed on a mobile phone.

Methodology

The system consists of:

- **10 tilt sensors** to detect finger bending
- **3-axis accelerometer** to capture hand movement
- **Arduino microcontroller** to process sensor data
- **Bluetooth module** to send recognized gestures to a smartphone

The recognition algorithm compares sensor values with stored gesture patterns and outputs the corresponding text.

Dataset

The experiment used gestures from **Malaysian Sign Language (MSL)** including:

- Alphabets: A, B, C
- Numbers: 1, 2, 3
- Words: “Saya”, “Makan”, “Apa”

Results

Experiments were conducted with **four participants** performing gestures ten times each.

Accuracy results:

- Alphabet recognition: **~95%**
- Number recognition: **~93.33%**
- Gesture recognition: **~78.33%**

Overall system accuracy was approximately **89%**.

Significance

The proposed system provides a **low-cost and practical communication solution** for deaf and speech-impaired people by converting sign language into text in real time.

Future Work

Future improvements suggested include:

- Increasing the number of gestures in the database
- Using advanced recognition models such as **Hidden Markov Models**
- Improving gesture recognition accuracy through larger training datasets.